

Chapter 4

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5.Find the value of the variables that make each product term 1 and each sum term 0.

(b) $\overline{A}\overline{B}C$ (d) $\overline{A} + B + \overline{C}$ (f) $\overline{A} + B$

Answer Area:

(b) A : 1, B : 0, C : 1;

(d) A : 1, B : 0, C : 1;

(f) A : 1, B : 0

6.Find the value of X for all possible values of variables:

(b) $X = \overline{(A + B)}C$ (d) $X = (A + B)(\overline{A} + B)$ (e) $X = (A + BC)(\overline{B} + \overline{C})$

Answer Area:

(b)

A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

(d)

A	B	X
0	0	0
0	1	1
1	0	0
1	1	1

(e)

A	B	C	X
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

8.Identity the Boolean rule(s) on which each of the following equalities is based

(b) $A\overline{A}B + AB\overline{C} + AB\overline{B} = AB\overline{C}$

(d) $AB(C + \overline{C}) + AC = AB + AC$

(f) $ABC + \overline{A}\overline{B} + \overline{A}BC\overline{D} = ABC + \overline{A}\overline{B} + D$

Answer Area:

- (b) Rule 8: $A\bar{A} = 0$, $0 \cdot B = 0$
 (d) Rule 6: $A + \bar{A} = 1$
 (f) Rule 11: $A + \bar{A}B = A + B$

10. Apply DeMorgan's theorems to each expression:

(b) $\overline{AB(CD + EF)}$

(d) $\overline{(\bar{A} + B + C + D)(\bar{A}\bar{B}\bar{C}\bar{D})}$

Answer Area:

(b)

$$\overline{AB(CD + EF)} = \overline{AB} + \overline{CD + EF} = (\bar{A} + \bar{B}) + \overline{CDEF} = \bar{A} + \bar{B} + (\bar{C} + \bar{D})(\bar{E} + \bar{F})$$

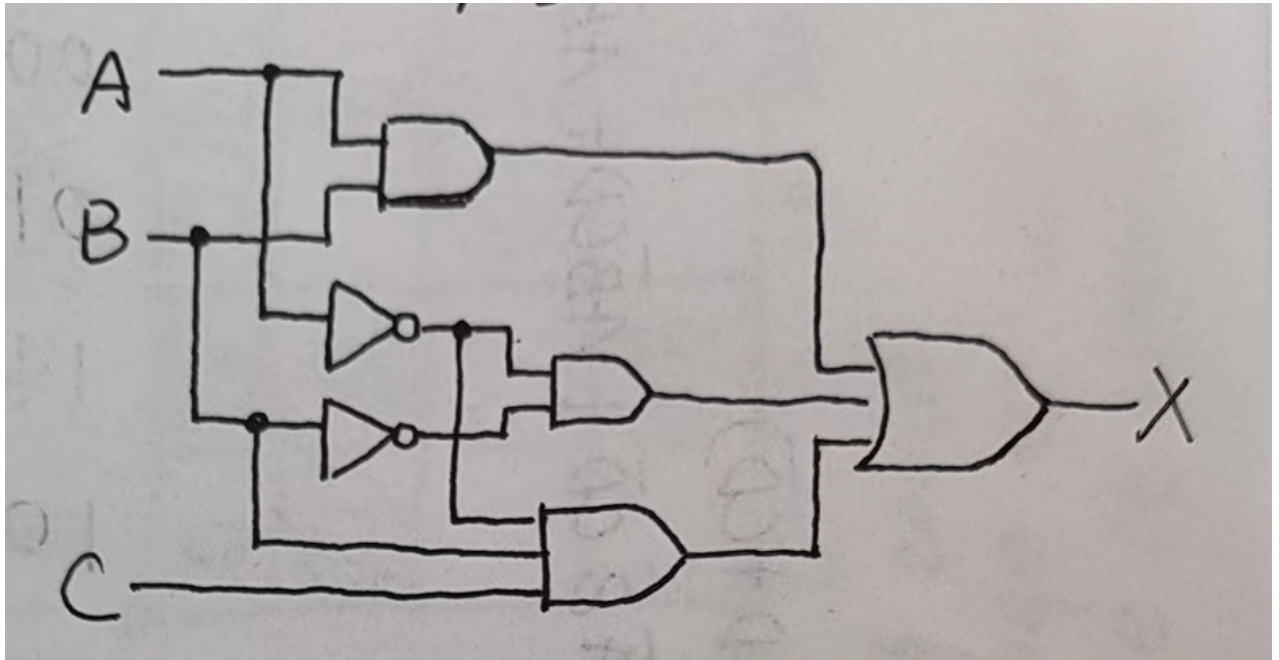
(d)

$$\overline{(\bar{A} + B + C + D) \cdot (\bar{A}\bar{B}\bar{C}\bar{D})} = \bar{A} + B + C + D + \bar{A}\bar{B} \cdot \bar{C}\bar{D}$$

15. Draw a logic circuit represented by each expression: (b) $AB + \bar{A} \cdot \bar{B} + \bar{A}BC$

Answer area:

(b)



18. Construct a truth table for each of the following Boolean expressions:

(c) $AB + BC$ (e) $(A + B)(\bar{B} + C)$

Answer Area:

(e)

A	B	C	$AB + BC$
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

(e)

A	B	C	$(A + B)(\overline{B} + C)$
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

21. Using Boolean algebra, simplify each expression:

(b) $\overline{A} \cdot \overline{B}C + (A + B + \overline{C}) + \overline{A} \cdot \overline{B} \cdot \overline{C}D$

(d) $ABCD + AB(\overline{C}D) + (\overline{A}B\overline{C}D)$

Answer Area:

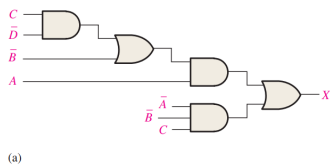
(b)

$$\begin{aligned}
 & \overline{A} \cdot \overline{B}C + \overline{(A + B + \overline{C})} + \overline{A} \cdot \overline{B} \cdot \overline{C}D \\
 = & \overline{A} \cdot \overline{B}C + \overline{A} \cdot \overline{B}C + \overline{A} \cdot \overline{B} \cdot \overline{C}D \\
 = & \overline{A} \cdot \overline{B}C + \overline{A} \cdot \overline{B} \cdot \overline{C}D \\
 = & \overline{A} \cdot \overline{B}(C + \overline{C}D) \\
 = & \overline{A} \cdot \overline{B}(C + D) \\
 = & \overline{A} \cdot \overline{B}C + \overline{A} \cdot \overline{B}D
 \end{aligned}$$

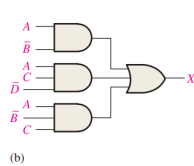
(d)

$$\begin{aligned}
 & ABCD + AB(\overline{C}D) + (\overline{A}B\overline{C}D) \\
 = & AB(CD + \overline{C}D) + (\overline{A}B)CD \\
 = & AB + (\overline{A}B)CD \\
 = & AB + CD
 \end{aligned}$$

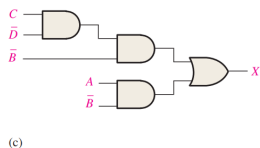
22. Determine which of the logic circuit in Figure 4-51 are equivalent:



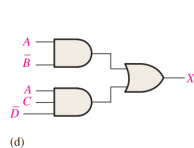
(a)



(b)



(c)



(d)

- (a) The circuit is equivalent to
 $(C\overline{D} + \overline{B})A + \overline{A} \cdot \overline{B}C = AC\overline{D} + A\overline{B} + \overline{B}C$
- (b) The circuit is equivalent to
 $A\overline{B} + AC\overline{D} + \overline{A}BC = A\overline{B} + AC\overline{D}$
- (c) The circuit is equivalent to
 $A\overline{B} + \overline{B}C\overline{D}$
- (d) The circuit is equivalent to
 $A\overline{B} + AC\overline{D}$
- Therefore, (b) and (d) are equivalent.

24. Convert the following expressions to sum-of-products(SOP) forms:

- (a) $AB + CD(A\overline{B} + CD)$ (b) $AB(\overline{B} \cdot \overline{C} + BD)$ (c) $A + B[AC + (B + \overline{C})D]$

Answer Area:

- (a)
 $AB + CD(A\overline{B} + CD)$
 $= AB + A\overline{B}CD + CDCD$
 $= AB + A\overline{B}CD + CD$
 $= AB + (A\overline{B} + 1)CD$
 $= AB + CD$
- (b)
 $AB(\overline{B} \cdot \overline{C} + BD)$
 $= A(B\overline{B})\overline{C} + ABBD$
 $= ABD$
- (c)
 $A + B[AC + (B + \overline{C})D]$
 $= A + ABC + BBD + B\overline{C}D$
 $= A(1 + BC) + BD(1 + \overline{C})$
 $= A + BD$

26. Convert each SOP expression in Problem 24 to standard SOP form.

Answer Area:

- (a)
 $AB + CD(A\overline{B} + CD)$
 $= AB + CD$
 $= AB(C + \overline{C})(D + \overline{D}) + (\overline{A} + A)(B + \overline{B})CD$
 $= ABCD + ABC\overline{D} + A\overline{B}CD + A\overline{B}C\overline{D} + \overline{A}BCD + \overline{A}B\overline{C}D + \overline{A} \cdot \overline{B}CD$
- (b)
 $AB(\overline{B} \cdot \overline{C} + BD)$
 $= ABD$
 $= ABD(C + \overline{C})$
 $= ABCD + AB\overline{C}D$
- (c)
 $A + B[AC + (B + \overline{C})D]$

$$\begin{aligned}
&= A + BD \\
&= A(B + \overline{B})(C + \overline{C})(D + \overline{D}) + (A + \overline{A})BD(C + \overline{C}) \\
&= ABCD + ABC\overline{D} + AB\overline{C}D + AB\overline{C}\cdot\overline{D} + A\overline{B}CD + A\overline{B}C\overline{D} + A\overline{B}\cdot\overline{C}D + A\overline{B}\cdot\overline{C}\cdot\overline{D} + \\
&\quad \overline{A}BCD + \overline{A}B\overline{C}D
\end{aligned}$$

33. Develop a truth table for each of the SOP expression:

(a) $\overline{A}B + AB\overline{C} + \overline{A}\cdot\overline{C} + A\overline{B}C$

Answer Area:

A	B	C	$\overline{A}B + AB\overline{C} + \overline{A}\cdot\overline{C} + A\overline{B}C$
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

35. Develop a truth table for each of the standard POS expressions::

(b) $(A + \overline{B})(A + \overline{B} + \overline{C})(B + C + \overline{D})(\overline{A} + B + \overline{C} + D)$

Answer Area:

(b)

A	B	C	D	$(A + \overline{B})(A + \overline{B} + \overline{C})(B + C + \overline{D})(\overline{A} + B + \overline{C} + D)$
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

36. For each truth table in Table 4-14, derive a standard SOP and a standard POS expression.

(a)

A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

(c)

A	B	C	D	X
0	0	0	0	1
0	0	0	1	1
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	0
1	0	0	0	0
1	0	0	1	1
1	0	1	0	0
1	0	1	1	0
1	1	0	0	1
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

Answer Area:

(a) SOP: $\overline{A} \cdot \overline{B}C + A\overline{B} \cdot \overline{C} + A\overline{B}C + ABC$

POS: $(A + B + C)(A + \overline{B} + C)(A + \overline{B} + \overline{C})(\overline{A} + \overline{B} + C)$

(c) SOP: $\overline{A} \cdot \overline{B} \cdot \overline{C} \cdot \overline{D} + \overline{A} \cdot \overline{B} \cdot \overline{C}D + \overline{A} \cdot \overline{B}CD + \overline{A}B\overline{C}D + \overline{A}BC\overline{D} + A\overline{B} \cdot \overline{C}D + ABC \cdot \overline{D}$

POS: $(A + B + \overline{C} + D)(A + \overline{B} + C + D)(A + \overline{B} + \overline{C} + \overline{D})(\overline{A} + B + C + D)(\overline{A} + B + \overline{C} + D)(\overline{A} + B + \overline{C} + \overline{D})(\overline{A} + \overline{B} + C + D)(\overline{A} + \overline{B} + \overline{C} + D)(\overline{A} + \overline{B} + \overline{C} + \overline{D})$

40. Use a Karnaugh map to find the minimum SOP form for each expression:

(c) $\overline{A}(BC + B\overline{C}) + A(BC + B\overline{C})$

Answer Area:

(c)

$A \backslash BC$	00	01	11	10
0			1	1
1			1	1

$$\bar{A}(BC + B\bar{C}) + A(BC + B\bar{C})$$

$$= \bar{A}BC + \bar{A}B\bar{C} + ABC + AB\bar{C}$$

$$= B$$

44. Use a Karnaugh map to reduce each expression to a minimum SOP form:

(c) $\bar{A}B(\bar{C} \cdot \bar{D} + \bar{C}D) + AB(\bar{C} \cdot \bar{D} + \bar{C}D) + \bar{A}\bar{B} \cdot \bar{C}D$

(d) $(\bar{A} \cdot \bar{B} + \bar{A}\bar{B})(CD + C\bar{D})$

Answer Area:

(c)

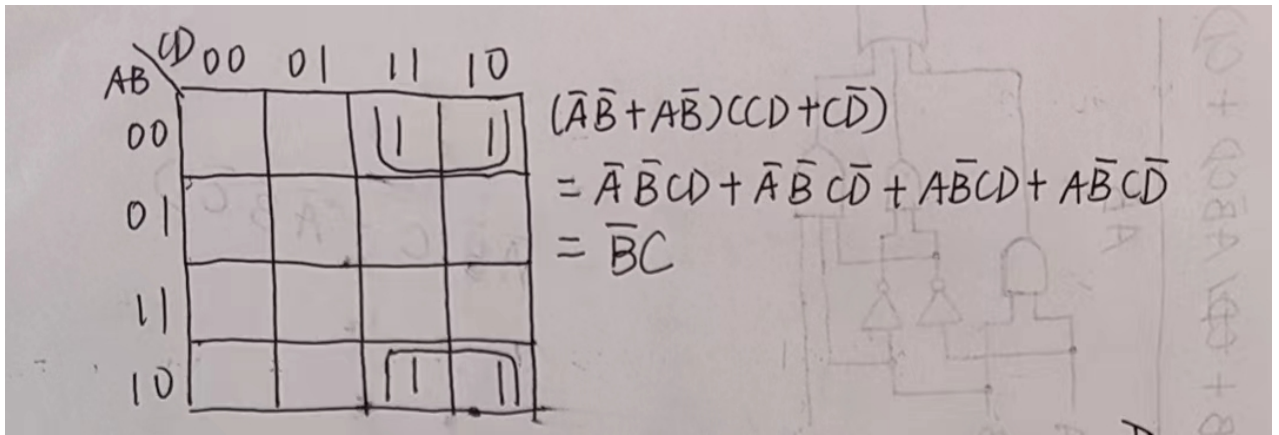
$AB \backslash CD$	00	01	11	10
00				1
01	1	1		
11	1	1		
10		1		

$$\bar{A}B(\bar{C}\bar{D} + \bar{C}D) + ABC\bar{C}\bar{D} + \bar{C}D + \bar{A}\bar{B}\bar{C}D$$

$$= \bar{A}B\bar{C}\bar{D} + \bar{A}B\bar{C}D + ABC\bar{C}\bar{D} + ABC\bar{C}D + \bar{A}\bar{B}\bar{C}D$$

$$= B\bar{C} + A\bar{C}D$$

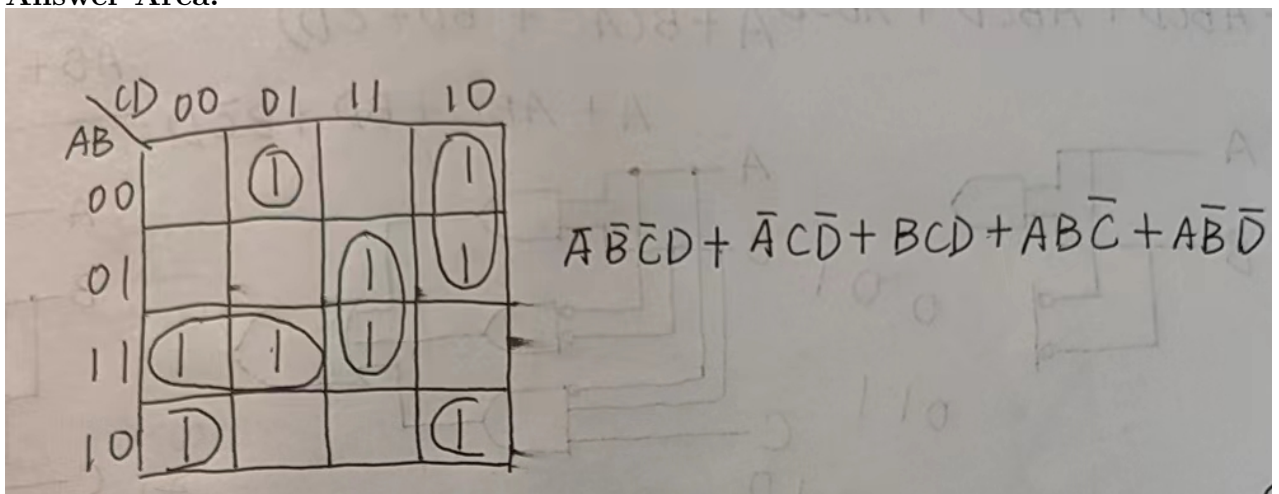
(d)



46. Use the Karnaugh map method to implement the minimum SOP expression for the logic function specified in truth Table 4-16.

A	B	C	D	X
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	0
1	1	1	1	1

Answer Area:



48. Use a Karnaugh map to find the minimum POS for each expression:

(b) $(X + \bar{Y})(\bar{X} + Z)(X + \bar{Y} + \bar{Z})(\bar{X} + \bar{Y} + Z)$

Answer Area:

$(X + \bar{Y})(\bar{X} + Z)(X + \bar{Y} + \bar{Z})(\bar{X} + \bar{Y} + Z)$
 $= (X + \bar{Y})(\bar{X} + Z)$

49. Use a Karnaugh map to simplify each expression to minimum POS form.

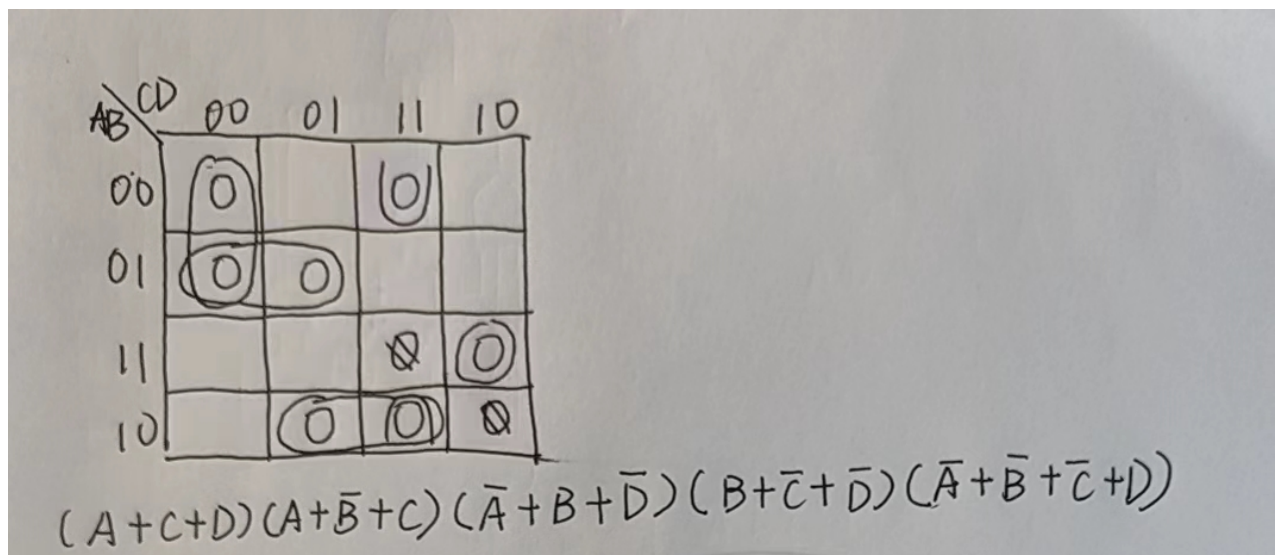
(a) $(A + \bar{B} + C + \bar{D})(\bar{A} + B + \bar{C} + D)(\bar{A} + \bar{B} + \bar{C} + \bar{D})$

Answer Area:

Therefore, $(A + \bar{B} + C + \bar{D})(\bar{A} + B + \bar{C} + D)(\bar{A} + \bar{B} + \bar{C} + \bar{D})$ is the simplest POS form.

51. Determine the minimum POS expression for the function in Table 4-16.

Answer Area:



Part 2: Give the simplest SOP form for segment f.

Answer Area:

A	B	C	D	f
0	0	0	0	1
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	x
1	0	1	1	x
1	1	0	0	x
1	1	0	1	x
1	1	1	0	x
1	1	1	1	x

AB \ CD	00	01	11	10
00	1			
01	1	1		1
11	X	X	X	X
10	1	1	X	X

$$f = \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}B\bar{C}\bar{D} + \bar{A}B\bar{C}D + \bar{A}BC\bar{D} + A\bar{B}\bar{C}\bar{D} + A\bar{B}\bar{C}D = A + \bar{C}\bar{D} + B\bar{C} + B\bar{D}$$

